Sensor README

Contents

1.1 Sensors data files description 1
1.1.1 Naming conventions 1
1.1.2 Data structures 2
1.1.2.1 Radar data 2
1.1.2.2 Radiofrequency (RF) direction finding (DF) sensor data 6
1.1.2.3 EO/IR data 9
1.1.2.4 Acoustic data 11
1.1.2.5 Cyber data 11
1.1.2.6 Other 13
1.1.3 Classification types 15

1.1 Sensors data files description

This section will give a detailed description of the sensors data file structure. The sensor data reported in this section refers to: Radar data; Passive Radio Frequency (RF) Direction Finding data; EO/IR data; Acoustic Data; Cyber data; Other Sensors data.

1.1.1 Naming conventions

Each sensor data file is a .csv file reporting the naming convention:

<Team>_<Id>_<Type>_<YYMMDD(start)>_<HHMM(start)>_<HHMM(end)>.csv

Explanation of each entry of the sensor data file name is as follows:

- <Team> represents the team that operated the sensor. It is one of: Alpha, Bravo, Charlie, Delta, Echo.
- <Id> is the 4-letter Id of the sensor, which corresponds to the name of the sensor. The name is mapped to a random Id such that the participants do not have access to the manufacturer of the sensor.
- <Type> is the type of the sensor.
- <YYMMDD(takeoff)> - Date (year, month, day [UTC]) at which the sensor detection started.
- <HHMM(takeoff)> - Time (hours and minutes [UTC]) at which the sensor detection started.
- <HHMM(landing)> - Time (hours and minutes [UTC]) at which the sensor detection ended. The sensor detection start and end days are the same.

Sensor data file name example: *"Alpha_AAAA_CAMERA_240916_1300_1340.csv"*

1.1.2 Data structures

The data in the Sensors .csv files contains detection report messages. These reports contain information about the location of the sensor which performed the detection, as well as the location and classification of the detected objects. These information are provided as comma separated files with self-explanatory column names. Detailed structure description for each of the sensor is reported below.

1.1.2.1 Radar data

These files give the outputs from different radar systems which detected the UAS flights. Data structure for each specific radar sensor is reported below as well as a brief description of each data entry. Each sensor name has been anonymised and distinguished with a unique id consisting of four letters.

i. IUKL

- *messageType:* The type of SAPIENT message report.
- *timestamp [UTC]:* The recorded time of the message report.
- *nodeId:* Unique identifier for the sensor node generating the report.
- *reportId:* Identifier for the specific report.
- *sensor_location_x [°]:* The geographic longitude coordinate of the sensor location.
- *sensor_location_y [°]:* The geographic latitude coordinate of the sensor location.
- *sensor_location_z [m]:* Altitude of the sensor.
- *sensor_location_coordinateSystem:* Coordinate system used to define the sensor's location.
- *sensor_location_datum:* Reference datum used for the sensor location.
- *location_x [°]:* The geographic longitude coordinate of the detected object's location.
- *location_y [°]:* The geographic latitude coordinate of the detected object's location.
- *location_z [m]:* Altitude of the detected object.
- *location_xError [m]:* Estimated error in the X-direction (longitude coordinate) of the detected object's location.
- *location_yError [m]:* Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- *location_zError [m]:* Estimated error in the Z-direction (altitude) of the detected object's location.
- *location_coordinateSystem:* Coordinate system used to define the detected object's location.
- *location_datum:* Reference datum used for the detected object's location.
- *classification:* List of possible categorizations of the detected object.
- *trackInfo*: Additional information about the detected object's track.
- *objectId:* Identifier for the object being detected.
- *taskId:* Identifier for the task.
- *state:* Current operational or behavioural state of the detected object (e.g., active, lost).

ii. UXYJ

- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_xError [m]: Estimated error in the X-direction (longitude coordinate) of the detected object's location.
- location_yError [m]: Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- location_zError [m]: Estimated error in the Z-direction (altitude) of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- detectionConfidence: Confidence score (between 0 and 1) indicating reliability of the detection.
- enuVelocity_eastRate [m/s]: Detected object's velocity component toward East in ENU coordinates.
- enuVelocity_northRate [m/s]: Detected object's velocity component toward North in ENU coordinates.
- enuVelocity_upRate [m/s]: Detected object's upward velocity component.
- enuVelocity_eastRateError [m/s]: Estimated error in eastward velocity.
- enuVelocity_northRateError [m/s]: Estimated error in northward velocity.
- enuVelocity_upRateError [m/s]: Estimated error in upward velocity.
- enuVelocity_eastNorthRateUnits: Measurement units used for East and North velocity components.
- enuVelocity_upRateUnits: Measurement units for the upward velocity component.
- objectId: Identifier for the object being detected.
- taskId: Identifier for the task.

iii. EGOI

- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.

- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_xError [m]: Estimated error in the X-direction (longitude coordinate) of the detected object's location.
- location_yError [m]: Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- location_zError [m]: Estimated error in the Z-direction (altitude) of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- detectionConfidence: Confidence score (between 0 and 1) indicating reliability of the detection.
- enuVelocity_eastRate [m/s]: Detected object's velocity component toward East in ENU coordinates.
- enuVelocity_northRate [m/s]: Detected object's velocity component toward North in ENU coordinates.
- enuVelocity_upRate [m/s]: Detected object's upward velocity component.
- enuVelocity_eastNorthRateUnits: Measurement units used for East and North velocity components.
- enuVelocity_upRateUnits: Measurement units for the upward velocity component.
- objectInfo: Additional information or description about the object being tracked.
- objectId: Identifier for the object being detected.
- taskId: Identifier for the task.

iv. NYTF

- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- detectionConfidence: Confidence score (between 0 and 1) indicating reliability of the detection.
- enuVelocity_eastRate [m/s]: Detected object's velocity component toward East in ENU coordinates.

- enuVelocity_northRate [m/s]: Detected object's velocity component toward North in ENU coordinates.
- enuVelocity_upRate [m/s]: Detected object's upward velocity component.
- enuVelocity_eastNorthRateUnits: Measurement units used for East and North velocity components.
- enuVelocity_upRateUnits: Measurement units for the upward velocity component.
- trackInfo: Additional information about the detected object's track.
- objectId: Identifier for the object being detected.
- taskId: Identifier for the task.
- state: Current operational or behavioral state of the detected object (e.g., active, lost)

v. EUIG
- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- classification: List of possible categorizations of the detected object.
- detectionConfidence: Confidence score (between 0 and 1) indicating reliability of the detection.
- enuVelocity_eastRate [m/s]: Detected object's velocity component toward East in ENU coordinates.
- enuVelocity_northRate [m/s]: Detected object's velocity component toward North in ENU coordinates.
- enuVelocity_upRate [m/s]: Detected object's upward velocity component.
- enuVelocity_eastNorthRateUnits: Measurement units used for East and North velocity components.
- enuVelocity_upRateUnits: Measurement units for the upward velocity component.
- trackInfo: Additional information about the detected object's track.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.
- additionalInformation: Free-text field used for any supplementary data not captured in other columns.

1.1.2.2 Radiofrequency (RF) direction finding (DF) sensor data

These files give the outputs from different radiofrequency direction finding systems. Data structure for each specific RF-DF sensor is reported below as well as a brief description of each data entry. Each sensor name has been anonymised and distinguished with a unique id consisting of four letters.

i. ILYS

- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_xError [m]: Estimated error in the X-direction (longitude coordinate) of the detected object's location.
- location_yError [m]: Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- location_zError [m]: Estimated error in the Z-direction (altitude) of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.
- taskId: Identifier for the task.

ii. MMJJ

- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_xError [m]: Estimated error in the X-direction (longitude coordinate) of the detected object's location.

- location_yError [m]: Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- location_zError [m]: Estimated error in the Z-direction (altitude) of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.

iii. ORFH

- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- rangeBearing_range [m]: Measured distance from sensor to detected object.
- rangeBearing_azimuth [rad]: Angle of the detected object relative to North, based on sensor's concept of North.
- rangeBearing_coordinateSystem: Coordinate system used for range and bearing values.
- rangeBearing_datum: Reference datum used for range and bearing values.
- classification: List of possible categorizations of the detected object.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.
- destinationId: Identifier of the node towards which the current report has been sent.

iv. SMFS

- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.

- classification: List of possible categorizations of the detected object.
- objectId: Identifier for the object being detected.
- behaviour: Inferred or observed behavior of the detected object (e.g., moving, stationary).

v. IKJX
- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor .
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_xError [m]: Estimated error in the X-direction (longitude coordinate) of the detected object's location.
- location_yError [m]: Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- location_zError [m]: Estimated error in the Z-direction (altitude) of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.

vi. OEPS:
- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- rangeBearing_azimuth [rad]: Angle of the detected object relative to North, based on sensor's concept of North.
- rangeBearing_azimuthError [rad]: Estimated error in azimuth measurement of the detected object.
- rangeBearing_coordinateSystem: Coordinate system used for range and bearing values.
- rangeBearing_datum: Reference datum used for range and bearing values.

- objectInfo: Additional information or description about the object being tracked.
- objectId: Identifier for the object being detected.
- signal: List of signals detected as part of this detection.

vii. GTYT:
- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- rangeBearing_azimuth [rad]: Angle of the detected object relative to North, based on sensor's concept of North.
- rangeBearing_azimuthError [rad]: Estimated error in azimuth measurement of the detected object.
- rangeBearing_coordinateSystem: Coordinate system used for range and bearing values.
- rangeBearing_datum: Reference datum used for range and bearing values.
- objectInfo: Additional information or description about the object being tracked.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.
- signal: List of signals detected as part of this detection.

1.1.2.3 EO/IR data

These files give the outputs from different Electro Optical (EO)/Infrared camera systems. Data structure for each specific EO/IR sensor is reported below as well as a brief description of each data entry. Each sensor name has been anonymised and distinguished with a unique id consisting of four letters.

i. WCFX:
- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- rangeBearing_range [m]: Measured distance from sensor to detected object.
- rangeBearing_azimuth [rad]: Angle of the detected object relative to North, based on sensor's concept of North.

- rangeBearing_azimuthError [rad]: Estimated error in azimuth measurement of the detected object.
- rangeBearing_coordinateSystem: Coordinate system used for range and bearing values.
- rangeBearing_datum: Reference datum used for range and bearing values.
- classification: List of possible categorizations of the detected object.
- trackInfo: Additional information about the detected object's track.
- objectInfo: Additional information or description about the object being tracked.
- objectId: Identifier for the object being detected.
- taskId: Identifier for the task.
- destinationId: Identifier of the node towards which the current report has been sent.
- state: Current operational or behavioral state of the detected object (e.g., active, lost).
- associatedDetection: In a fused detection, the list of detections that were used to generate the current detection.
- derivedDetection: A list of fused detections derived from the current detection.

ii. NJQL:

- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- rangeBearing_range [m]: Measured distance from sensor to detected object.
- rangeBearing_azimuth [rad]: Angle of the detected object relative to North, based on sensor's concept of North.
- rangeBearing_elevation [rad]: Angle of the detected object relative to the horizontal plane.
- rangeBearing_coordinateSystem: Coordinate system used for range and bearing values.
- classification: List of possible categorizations of the detected object.
- objectId: Identifier for the object being detected.
- taskId: Identifier for the task.

iii. FGIX:

- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- rangeBearing_range [m]: Measured distance from sensor to detected object.

- rangeBearing_azimuth [rad]: Angle of the detected object relative to North, based on sensor's concept of North.
- rangeBearing_elevation [rad]: Angle of the detected object relative to the horizontal plane.
- rangeBearing_coordinateSystem: Coordinate system used for range and bearing values.
- rangeBearing_datum: Reference datum used for range and bearing values.
- classification: List of possible categorizations of the detected object.
- detectionConfidence: Confidence score (between 0 and 1) indicating reliability of the detection.
- objectId: Identifier for the object being detected.
- taskId: Identifier for the task.
- destinationId: Identifier of the node towards which the current report has been sent.

1.1.2.4 Acoustic data

These files give the outputs from acoustic sensor data. Data structure for each specific acoustic sensor is reported below as well as a brief description of each data entry. Each sensor name has been anonymised and distinguished with a unique id consisting of four letters.

i. EMXGA:
- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- rangeBearing_range [m]: Measured distance from sensor to detected object.
- rangeBearing_azimuth [rad]: Angle of the detected object relative to North, based on sensor's concept of North.
- rangeBearing_coordinateSystem: Coordinate system used for range and bearing values.
- rangeBearing_datum: Reference datum used for range and bearing values.
- classification: List of possible categorizations of the detected object.
- objectId: Identifier for the object being detected.
- destinationId: Identifier of the node towards which the current report has been sent.
- behaviour: Inferred or observed behavior of the detected object (e.g., moving, stationary).

1.1.2.5 Cyber data

These files give the outputs from different cyber systems. Data structure for each specific cyber sensor is reported below as well as a brief description of each data entry. Each sensor name has been anonymized and distinguished with a unique id consisting of four letters.

a) CEQX:
- messageType: The type of SAPIENT message report.

- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_xError [°]: Estimated error in the X-direction (longitude coordinate) of the detected object's location.
- location_yError [°]: Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- location_zError [m]: Estimated error in the Z-direction (altitude) of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- enuVelocity_eastRate [m/s]: Detected object's velocity component toward East in ENU coordinates.
- enuVelocity_northRate [m/s]: Detected object's velocity component toward North in ENU coordinates.
- enuVelocity_upRate [m/s]: Detected object's upward velocity component.
- enuVelocity_eastRateError [m/s]: Estimated error in eastward velocity.
- enuVelocity_northRateError [m/s]: Estimated error in northward velocity.
- enuVelocity_upRateError [m/s]: Estimated error in upward velocity.
- enuVelocity_eastNorthRateUnits: Measurement units used for East and North velocity components.
- enuVelocity_upRateUnits: Measurement units for the upward velocity component.
- trackInfo: Additional information about the detected object's track.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.
- taskId: Identifier for the task.
- destinationId: Identifier of the node towards which the current report has been sent.
- state: Current operational or behavioral state of the detected object (e.g., active, lost).

b) STUG:
   - messageType: The type of SAPIENT message report.
   - timestamp [UTC]: The recorded time of the message report.
   - nodeId: Unique identifier for the sensor node generating the report.
   - reportId: Identifier for the specific report.
   - sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
   - sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
   - sensor_location_z [m]: Altitude of the sensor.
   - sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
   - sensor_location_datum: Reference datum used for the sensor location.

- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_xError [m]: Estimated error in the X-direction (longitude coordinate) of the detected object's location.
- location_yError [m]: Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- location_zError [m]: Estimated error in the Z-direction (altitude) of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- detectionConfidence: Confidence score (between 0 and 1) indicating reliability of the detection.
- enuVelocity_eastRate [m/s]: Detected object's velocity component toward East in ENU coordinates.
- enuVelocity_northRate [m/s]: Detected object's velocity component toward North in ENU coordinates.
- enuVelocity_eastNorthRateUnits: Measurement units used for East and North velocity components.
- objectInfo: Additional information or description about the object being tracked.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.
- taskId: Identifier for the task.
- state: Current operational or behavioral state of the detected object (e.g., active, lost).
- behaviour: Inferred or observed behavior of the detected object (e.g., moving, stationary).
- associatedDetection: In a fused detection, the list of detections that were used to generate the current detection.

1.1.2.6 Other

These files give the outputs from different UTM (i.e. Unmanned Traffic Management) systems. Data structure for each specific UTM sensor is reported below as well as a brief description of each data entry. Each sensor name has been anonymized and distinguished with a unique id consisting of four letters

a) VKYY
   - messageType: The type of SAPIENT message report.
   - timestamp [UTC]: The recorded time of the message report.
   - nodeId: Unique identifier for the sensor node generating the report.
   - reportId: Identifier for the specific report.
   - sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
   - sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
   - sensor_location_z [m]: Altitude of the sensor.
   - sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
   - sensor_location_datum: Reference datum used for the sensor location.
   - location_x [°]: The geographic longitude coordinate of the detected object's location.
   - location_y [°]: The geographic latitude coordinate of the detected object's location.
   - location_z [m]: Altitude of the detected object.

- location_xError [m]: Estimated error in the X-direction (longitude coordinate) of the detected object's location.
- location_yError [m]: Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- location_zError [m]: Estimated error in the Z-direction (altitude) of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- detectionConfidence: Confidence score (between 0 and 1) indicating reliability of the detection.
- enuVelocity_eastRate [m/s]: Detected object's velocity component toward East in ENU coordinates.
- enuVelocity_northRate [m/s]: Detected object's velocity component toward North in ENU coordinates.
- enuVelocity_upRate [m/s]: Detected object's upward velocity component.
- enuVelocity_eastNorthRateUnits: Measurement units used for East and North velocity components.
- enuVelocity_upRateUnits: Measurement units for the upward velocity component.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.

b) QTVS
- messageType: The type of SAPIENT message report.
- timestamp [UTC]: The recorded time of the message report.
- nodeId: Unique identifier for the sensor node generating the report.
- reportId: Identifier for the specific report.
- sensor_location_x [°]: The geographic longitude coordinate of the sensor location.
- sensor_location_y [°]: The geographic latitude coordinate of the sensor location.
- sensor_location_z [m]: Altitude of the sensor.
- sensor_location_coordinateSystem: Coordinate system used to define the sensor's location.
- sensor_location_datum: Reference datum used for the sensor location.
- location_x [°]: The geographic longitude coordinate of the detected object's location.
- location_y [°]: The geographic latitude coordinate of the detected object's location.
- location_z [m]: Altitude of the detected object.
- location_xError [m]: Estimated error in the X-direction (longitude coordinate) of the detected object's location.
- location_yError [m]: Estimated error in the Y-direction (latitude coordinate) of the detected object's location.
- location_zError [m]: Estimated error in the Z-direction (altitude) of the detected object's location.
- location_coordinateSystem: Coordinate system used to define the detected object's location.
- location_datum: Reference datum used for the detected object's location.
- classification: List of possible categorizations of the detected object.
- detectionConfidence: Confidence score (between 0 and 1) indicating reliability of the detection.

- enuVelocity_eastRate [m/s]: Detected object's velocity component toward East in ENU coordinates.
- enuVelocity_northRate [m/s]: Detected object's velocity component toward North in ENU coordinates.
- enuVelocity_upRate [m/s]: Detected object's upward velocity component.
- enuVelocity_eastNorthRateUnits: Measurement units used for East and North velocity components.
- enuVelocity_upRateUnits: Measurement units for the upward velocity component.
- id: Unique identifier for the current data entry.
- objectId: Identifier for the object being detected.

1.1.3 Classification types

The following reported classification types can be found in the sensor data files ‘classification’, ‘objectInfo’ as well as ‘trackInfo’ entries:

- ‘Air Vehicle',
- 'UAV',
- 'UAV Fixed Wing',
- 'UAV Rotary Wing',
- 'UAV Transmitter',
- 'Manned Fixed Wing',
- 'Manned Rotary Wing',
- 'Animal',
- 'Bird',
- 'EM Transmitter',
- 'FIXED_WING',
- 'Fixed Object',
- 'Home',
- 'Land Vehicle',
- 'Sea Vessel',
- 'Other',
- 'Unknown'